

ELEC-H415 Communication channels

Channel modeling with ray-tracing

1. Objective

The objective of this project is to develop a ray tracing code to calculate the propagation of electromagnetic waves, enabling the calculation of the power received by a user equipment (UE) connected to a 5G small cell base station (BS). This simulator will also determine the BS coverage area and the bit rate based on the UE location.

2. Hypotheses

Ray-tracing software allows us to model the propagation of electromagnetic waves under far-field assumptions. The waves are assumed to be locally plane, and each of their interactions with the environment can be calculated using the models covered in the course (reflection, transmission).

Our approach will be purely two-dimensional: the BS and UE are located at the same height above the ground, and only waves propagating in the horizontal plane are taken into consideration. It is not necessary to take ground reflection into account initially, but it could easily be added later, as a bonus (see 'morning workout').

We consider a BS operating at 26 GHz, with a bandwidth $B = 200$ MHz. The transmit and receive antennas are lossless vertical $\lambda/2$ dipoles. The transmit power is 20 dBm. We suppose that all materials in the environment have a real relative permittivity $\epsilon_r = 4$.

Each wave can undergo several successive interactions throughout its path. We only consider the following waves:

- The direct wave.
- Waves that have undergone up to three reflections (generalization to an arbitrary number of reflections can be added later, by using recursivity, as a bonus)
- No diffraction is considered, due to the high operating frequency.
- Each wave may also have been transmitted through walls, but transmission through an entire building is negligible.

3. Base station coverage

To establish a connection, the received power must exceed the UE sensitivity. This threshold depends on the desired bit rate: the higher the bit rate, the higher the sensitivity. The area where the signal received from the BS exceeds the UE sensitivity is called the BS coverage area.

In this project, we assume that the UE sensitivity (in dBm) varies linearly with the maximum achievable bit rate when both quantities are expressed in a logarithmic scale. The extreme values of variation are:

Sensitivity	Bit rate
-80 dBm	100 Mb/s
-60 dBm	1 Gb/s

Below -80 dBm, communication is impossible. Above -60 dBm, the maximum data rate is capped.

4. Environment

Freely choose a (simplified) outdoor environment where you want to deploy your BS (street, campus...). Ray-tracing calculations can be quite long (several hours), so starting with a simple geometry could be a good idea ☺.

The thickness of the walls is not considered when defining the map of the environment, nor when drawing the rays (but it is used when calculating the reflection and transmission coefficients).

5. Results

- Validate your simulator using simple cases: free-space first, then single reflection (as we did for the over-the-ground model) ... **This step is very important because, without rigorous validation, the simulator cannot be considered reliable.** Your final grade will depend on the quality of this validation.
- Determine the propagation paths between the BS and UE, using the image method. Trace the ray paths using your simulator and verify that all paths have a physical origin.
- Split your environment into local areas of about 1 sqm (to be adjusted depending on the chosen environment).
- Calculate in each local area the average power $\langle P_{RX} \rangle$.
- Plot a map of $\langle P_{RX} \rangle$ and corresponding data rate per local area. Deduce the coverage area of your BS.
- Deduce from the power map a path loss model: path loss law and standard deviation around it (NB: Ideally, statistics of $\langle P_{RX} \rangle$ around the path loss law should be log-normal, as we have seen in the theory, but, here, due to the simple geometry and the limited number of rays, other statistics could be found).
- If relevant, optimize the coverage of the full area by using several BSs. When multiple BSs are present, it is assumed that the UE connects to the BS offering the highest power. Therefore, do not add up the received powers coming from several BSs !

Exceeding the minimum objectives stated above is undoubtedly a plus, and it will be taken into account in the final grade. Almost the entire course could be simulated with your software (including TDL impulse responses, transfer functions...). Be creative!

The goal of a channel model is to create a surrogate (simplified) model that can easily simulate wireless communications later on. Therefore, it's essential to keep in mind throughout this project that your aim should be to provide, for example, a colleague with the necessary tools to emulate realistic channels without having to resort to ray tracing or measurements themselves.

As you can imagine, there won't be a single, universally applicable model, as the environment is freely chosen and each student project will be different. To meet this challenge, it's important to maintain consistent communication between us throughout the model development process (weekly meetings) to discuss progress and collaboratively determine the most efficient approach.

Your report must be submitted via the ULB VU as a single PDF file containing the results and the documented code.